

Decode Labs Project 1

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- **Organization:** Decode Labs
- **Batch:** 2026

Objective

- Design a deterministic prompt to extract structured information from unstructured support tickets into JSON using Prompt Engineering techniques.

Problem Statement

- Automate the time-consuming manual processing of unstructured customer support messages by converting raw text into structured JSON format.

JSON Schema

- customer_name: string
- email: string
- user_id: string
- issue_type: string

Few-Shot Examples

Example 1: Login Issue

- **Input:** User U12345 (John Smith) unable to login.
- **Output:** { "customer_name": "JohnSmith", "issue_type": "login_issue" }

Example 2: Payment Issue

- **Input:** Sarah Wilson, payment failed during subscription.
- **Output:** { "customer_name": "Sarah Wilson", "issue_type": "payment_issue" }

Example 3: Password Reset

- **Input:** Mike Ross (U9999) forgot password.
- **Output:** { "customer_name": "Mike Ross", "issue_type": "password_reset" }

Final Prompt Rules

- Output valid JSON only; no markdown or explanations.
- Use only information present in the text; return null for missing data.
- Strict structure: Begin with { and end with }.

Prompt Screenshots

Prompt 1:

You are a data extraction engine.

Extract:

- customer_name
- email
- user_id
- issue_type

Output valid JSON only.

If information is missing return null.

###

Hello Team,

My name is John Smith.

My email is johnsmith@gmail.com.

I am unable to login to my account.

User ID: U12345

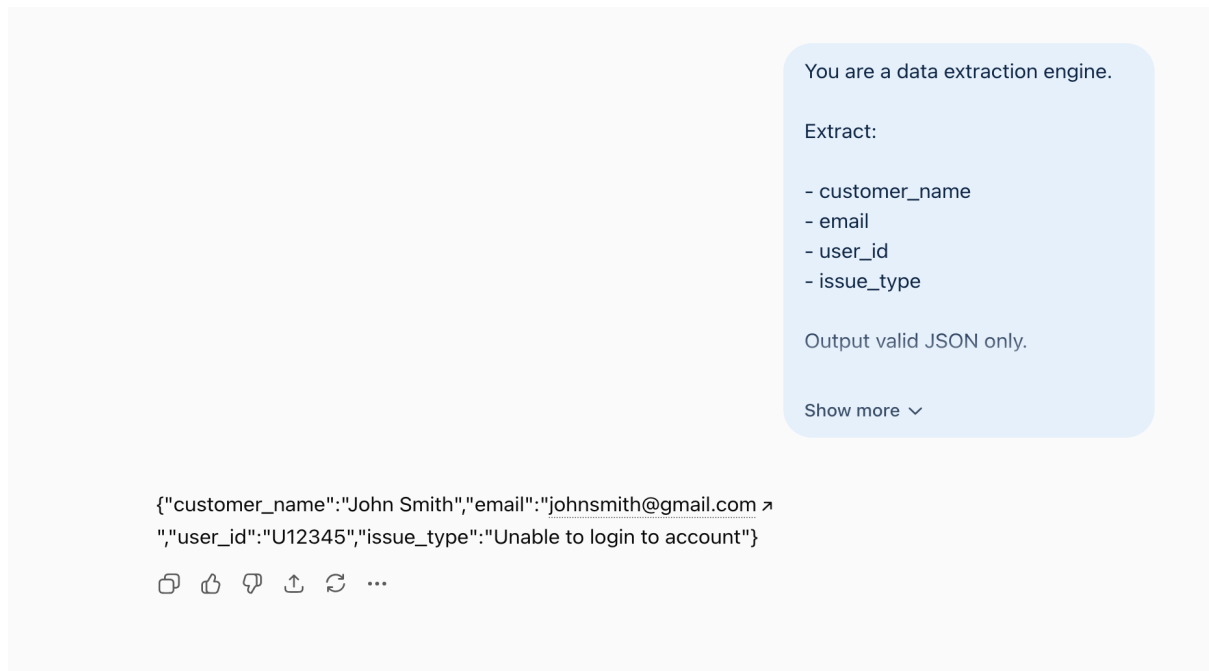
Please help.

Thanks,

John

###

Screenshot:



Prompt 2:

Hi Support,

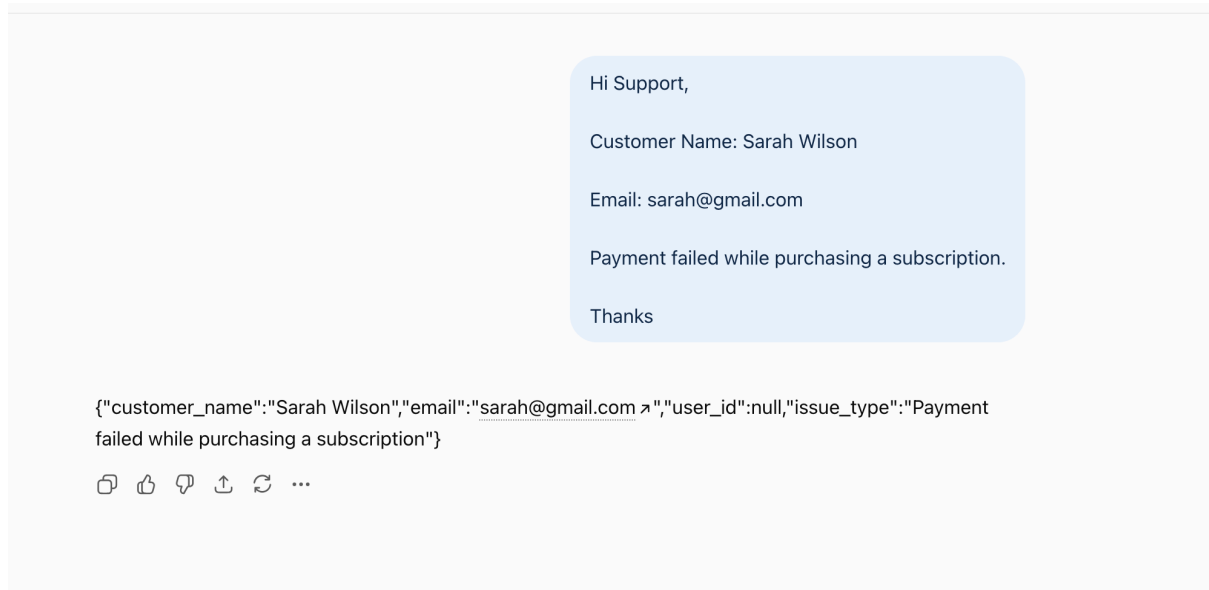
Customer Name: Sarah Wilson

Email: sarah@gmail.com

Payment failed while purchasing a subscription.

Thanks

Screenshot:



Prompt 3:

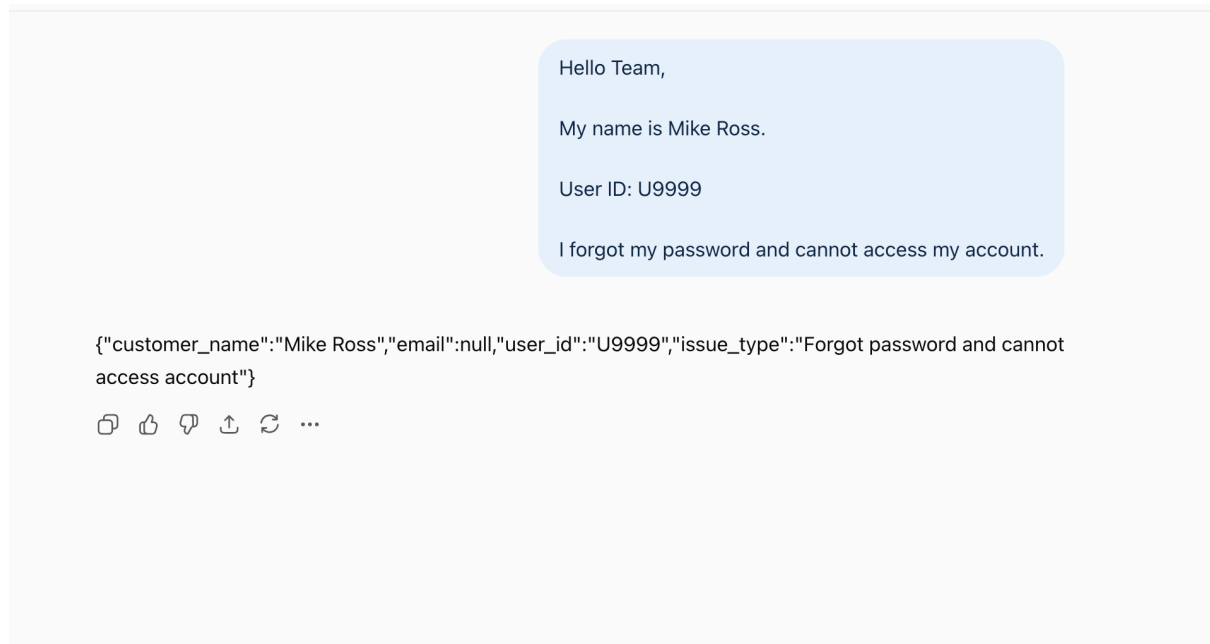
Hello Team,

My name is Mike Ross.

User ID: U9999

I forgot my password and cannot access my account.

Screenshot:



Results & Accuracy

- **Zero-Shot Accuracy:** 80%
- **Few-Shot Accuracy:** 98%

Conclusion

- Prompt Engineering (delimiters, few-shot, schema enforcement) effectively transforms unstructured data into consistent JSON outputs.

TOOLS USED

***ChatGPT**

***Claude**

***Google Docs**

***PDF export.**

